

MCS-221 Data Warehousing and Data Mining (2010–2023)

Examination Papers

Below we list **each question from all years (2010–2023)** (both June and December Term-End exams) followed by its complete answer. Identical or highly similar questions across years have been answered once and noted where repeated. Answers include theoretical explanations, definitions, examples, and references for clarity.

December 2021

Q1(a). Define “data warehouse” and state any four characteristics of data warehousing.

Answer: A **data warehouse (DW)** is a centralized repository designed for query and analysis, integrating data from multiple heterogeneous sources to support management decision-making. It is subject-oriented (organized by business subject, e.g. sales) and contains historical data ¹. Four key characteristics (often called the “4 S’s”) are:

- **Subject-oriented:** Organized around major subjects like customers, products, or sales, rather than application functions ¹.
- **Integrated:** Data from different sources are cleaned and standardized so there is a single, consistent representation (e.g. one common customer ID) ¹.
- **Time-variant:** Stores historical data (time-stamped) so trends over months or years can be analyzed (e.g. keeping all past sales, not just current) ¹.
- **Non-volatile:** Once entered, data in the warehouse is **not** changed. Data are loaded and read for analysis, but the historical records remain fixed ¹.

Example: A sales data warehouse might store customer purchase history year-by-year, integrating point-of-sale and online data into standardized tables for product, time and customer.

Q1(b). Explain MOLAP and HOLAP architecture (with diagrams or textual description).

Answer: **OLAP (Online Analytical Processing)** servers provide fast, multidimensional analysis.

- **MOLAP (Multidimensional OLAP):** Stores data in specialized **multidimensional cubes** (array-based storage). Aggregated data (pre-computed summaries) are stored in these cubes for fast querying. MOLAP is optimal for complex calculations and offers very fast query performance on the pre-summarized data ². Disadvantages include limited data volume (cubes can grow very large) and extra processing to rebuild cubes when data change ³.
- **HOLAP (Hybrid OLAP):** Combines MOLAP and ROLAP. Detail data resides in a relational database (ROLAP), while aggregates are stored in MOLAP cubes. HOLAP provides the speed of MOLAP (fast access at all aggregation levels) plus the scalability of ROLAP ⁴. Its complexity is higher because it must support both storage modes ⁵.

Textual schema: In a star or snowflake schema, a **MOLAP server** would store fact/dimension data in multidimensional format (fast summaries). A **HOLAP server** would keep detailed tables in a relational store and still maintain MOLAP-style cubes for aggregated views.

Q1(c). Define data cleaning. Why is noisy data a problem? How is noisy data handled during data cleaning?

Answer: Data cleaning (or cleansing) is the process of detecting and correcting (or removing) corrupt, inaccurate or irrelevant records from a dataset. This includes fixing typos, removing duplicates, and dealing with missing or outlier values. Noisy data (incorrect or random errors) can distort analysis by introducing false patterns. For example, a misplaced decimal or outlier can skew averages.

Handling noisy data: Common techniques include smoothing (binning, clustering) or transformation. *Binning* groups data into ranges and replaces values with bin averages. *Clustering-based cleaning* groups similar records and treats outliers differently. Other methods: removing outliers beyond a threshold, or using statistical methods (e.g. regression) to correct values. These steps improve data quality before mining or warehousing.

Q1(d). What is a decision tree? How is it useful in classification? Explain the process of constructing and representing a decision tree, with an example.

Answer: A decision tree is a flowchart-like tree structure used for predictive modeling. It is a **supervised learning** technique for classification (and regression) ⁶. Internally, each node represents a test on an attribute, branches are outcomes of the test, and each leaf node represents a class label (decision) ⁶.

- **Use in classification:** Decision trees partition the input space by feature tests to classify new examples. For instance, to predict loan approval, a tree might first check $\text{income} > X$, then credit score, and so on, ultimately assigning "Approve" or "Reject." They are intuitive and easy to visualize ⁶.
- **Construction process:** Starting at the root, select the attribute that best splits the training data (e.g. using information gain or Gini index). Partition the dataset by that attribute's values, creating child nodes. Repeat recursively for each child using the remaining attributes until stopping criteria (all examples in a node have the same class, or no attributes left). Each leaf then assigns the class.

Example: Suppose we build a tree to classify animals. The root test could be "Has feathers?" If yes, branch leads to "Bird" class; if no, next test "Is warm-blooded?", etc. The construction (e.g. using ID3 or CART) chooses splits based on which attribute best separates the animal classes in the training data. The final tree is a set of "if-then" rules representing the classification model.

Q2(a). What is text mining? Mention any three applications of text mining. Explain two text mining techniques with examples.

Answer: Text mining (or text data mining) is the process of extracting valuable information and patterns from unstructured text. It uses natural language processing (NLP), machine learning and statistics to transform raw text into structured data and discover insights ⁷.

- **Applications:** Sentiment analysis on social media to gauge customer opinion, document classification (e.g. spam detection), topic discovery in news, and recommendation (e.g. suggesting articles based on content).

- **Techniques:** *Summarization:* Automatically generates a concise summary of a document's main ideas ⁸. For example, news headline generation or abstracting long reports. Summarization can be extractive (select key sentences) or abstractive.
Sentiment analysis: Determines if text conveys positive or negative sentiment ⁹. For instance, analyzing product reviews to determine satisfaction. This typically uses polarity lexicons or ML classifiers on preprocessed text.

Other common tasks include tokenization (splitting text into words/tokens), part-of-speech tagging, and named-entity recognition, which all help in analyzing and structuring text data ¹⁰ ⁹.

Q2(b). *In mining multimedia data on the web, explain: (i) PageRank, (ii) HITS, (iii) Page Layout Analysis, (iv) Vision Page Segmentation.*

Answer:

- **PageRank:** An algorithm (used by Google) that ranks web pages by considering the number and quality of hyperlinks pointing to them. Pages with many high-quality inbound links rank higher. It treats each link as a "vote," but weighs votes by the pagerank of the linking page. Thus, a page linked from authoritative sites gets a higher rank ¹¹.
- **HITS (Hyperlink-Induced Topic Search):** An algorithm that identifies **hubs and authorities**. "Authorities" are pages with valuable content on a topic, and "hubs" are pages linking to many authorities. The algorithm iteratively computes hub and authority scores to rank pages.
- **Page Layout Analysis:** The process of interpreting the visual and structural layout of a webpage. This may involve analyzing HTML/CSS to determine the semantic structure (e.g. identifying headers, footers, main content regions). It helps in extracting content by distinguishing layout elements (like identifying which text is a title versus a side note).
- **Vision Page Segmentation:** A technique from computer vision applied to web pages (especially scanned or image-based pages) where the page is divided into regions (text blocks, images, ads). It mimics human page understanding to isolate meaningful content. For example, segmenting a news webpage image into headline, article text, and image areas using visual cues.

Q3(a). *Define a data lake. Explain the step-by-step process of creating a data lake.*

Answer: A **data lake** is a large storage repository that holds raw data (structured, semi-structured, and unstructured) in its native format, typically in a distributed file system. Unlike a data warehouse (which stores formatted data), a data lake stores data **"as is,"** waiting for future use ¹². It supports all data types (e.g. logs, images, JSON) and is designed for advanced analytics and machine learning.

Creating a data lake (high-level steps):

1. **Ingest data:** Collect data from diverse sources (transactional DBs, IoT sensors, logs, external APIs) into the lake storage (e.g. cloud object store). Ensure raw data is captured (often in files).
2. **Store persistently:** Keep original, immutable copies of all ingested data. This may involve partitioning by date or source.
3. **Catalog/Metadata:** Maintain metadata about the data (what it is, source, format) in a data catalog so users can find what they need.
4. **Cleansing/Transformation (optional):** For data to be usable, apply schema-on-read processing: when data is queried, transform or cleanse at that time. The lake itself holds raw data but may also store curated copies.
5. **Security and governance:** Implement access controls and data governance policies (e.g. who can read or modify which data) to keep the lake manageable.

6. **Provide tools:** Expose the data through query/analytics engines (like SQL query engines on data lake) and APIs so analysts and apps can process the data.

Example: To build a marketing data lake, one might ingest web clickstream logs, campaign data, and social media feeds into the lake. The raw logs are stored untouched. Analysts can then run transformation queries (e.g. via Spark) against the raw data to build reports, or refine data into a warehouse later.

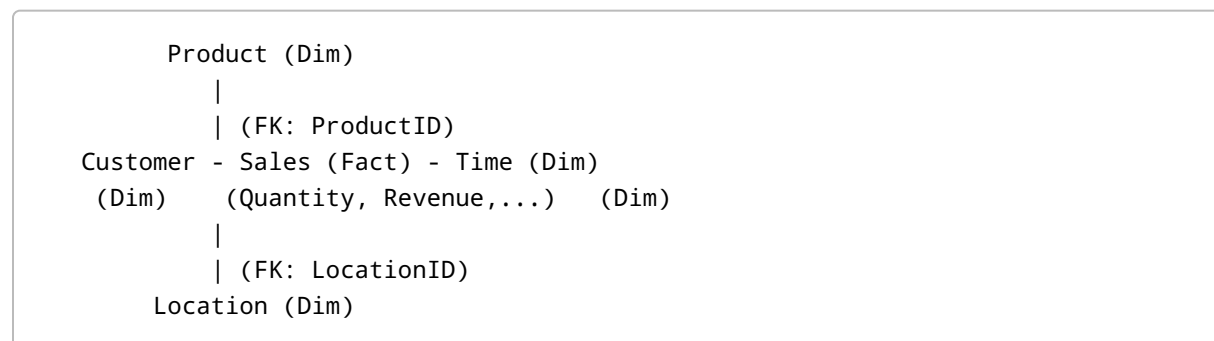
Q3(b). With an example, explain the star schema dimensional model.

Answer: In a **star schema**, a central *fact table* is connected to multiple *dimension tables*, resembling a star.

- **Fact table:** Contains numeric measurements (facts) of business processes, with foreign keys to dimensions. *Example:* A Sales fact table might have facts like `QuantitySold` and `Revenue`, and keys `ProductID`, `TimeID`, `StoreID`, `CustomerID`.
- **Dimension tables:** Contain descriptive attributes for each key. *Example:* `Product` dimension has `ProductID`, `Name`, `Category`, `Price`; `Customer` dimension has `CustomerID`, `Name`, `Region`, etc.; `Time` dimension has `TimeID`, `Day`, `Month`, `Year`; `Location` dimension has `LocationID`, `City`, `State`.

The fact table's foreign keys directly relate to primary keys of dimension tables. This flat structure (dimensions are not normalized further) makes queries simple and fast for aggregation.

Example diagram (textual):



Here, the Sales fact table has measures (Quantity, Amount) and foreign keys to `Product`, `Customer`, `Time`, `Location` dimensions. Each dimension table stores attributes describing that entity. Analysts can easily query like “total sales by Product category and Year” by joining fact to the relevant dims.

Q4(a). What is metadata? Explain its types and importance.

Answer: Metadata is “data about data” – information describing the structure, definitions, and usage of the data stored in the warehouse ¹³.

- **Types of metadata:**
- **Technical metadata:** Technical details needed to access/transform the data (e.g. schema, column definitions, data types, source systems, transformation rules) ¹³.

- **Business (or descriptive) metadata:** Contextual or business meaning of data (e.g. business definitions, data lineage, glossary terms, who owns the data) ¹³ .
- **Operational metadata:** Runtime information (e.g. when data was loaded, statistics about data volumes, job logs).
- **Importance:** Metadata enables users and systems to understand and effectively use the data warehouse. It tells users what each field means, where it came from, and how to query it. For example, metadata helps ensure that "SalesAmount" is understood as net of discounts, or that "CustomerID" links to a specific table. It also supports data governance (who changed data and when) and makes maintenance easier (e.g. knowing dependencies when updating tables).

Q4(b). List and describe the components of an ETL system. How do you improve ETL performance?

Answer: ETL (Extract, Transform, Load) is the process of moving data from source systems into the data warehouse. Its major components:

1. **Extraction:** Reading data from heterogeneous sources (databases, files, etc.). Tools connect to sources and pull data, often in batches or streams.
2. **Transformation:** Cleansing, enriching, and aggregating the extracted data. Includes data cleaning (handling missing values, noise), conforming data (standardizing formats), and business rule application. Data may be merged or summarized.
3. **Loading:** Writing the transformed data into the target warehouse tables (fact and dimension tables). This may involve loading into staging tables first and then into final schema.

Other components include: **Metadata repository** (managing ETL settings and rules), **data staging area** (temporary storage for raw or intermediate data), and **control module** (manages ETL workflows and scheduling).

Improving ETL performance:

- Use **incremental loads** (only new/changed data) instead of full reloads to reduce processing.
- **Parallelism:** Split tasks and run in parallel (e.g. parallel file reads, multi-threaded transformations).
- **Push-down processing:** Let the source database do filtering/aggregation to reduce data transferred.
- **Optimization:** Tune SQL queries and transformations; use bulk-load utilities.
- **Efficient staging:** Store data in optimized formats (e.g. columnar files), and clear staging after use.

Proper hardware (fast I/O, ample memory) and optimized ETL design (minimizing unnecessary transformations) also help speed up ETL processing.

Q5. Write short notes on any four of the following:

- (a) Data Marts
- (b) Generation of association rules
- (c) Apriori algorithm
- (d) Clustering and its methods
- (e) Bayes theorem in data mining.

Answer:

- **Data Mart:** A data mart is a subset of a data warehouse, focused on a specific business line or team (e.g.

sales, finance) ¹⁴ . It contains a portion of the DW data optimized for a particular group's needs. Data marts simplify data access for that group and can improve query performance by reducing data volume. Multiple data marts may exist within a larger DW.

- **Generation of Association Rules:** Association rule mining finds relationships (rules) between items in transactional data (e.g. market basket analysis). It has two steps: (1) find *frequent itemsets* that meet a minimum support threshold (items often bought together), and (2) generate rules from these itemsets that meet a minimum confidence threshold (e.g. if bread and milk frequently appear together, the rule might be “bread $\Rightarrow$ milk” with confidence equal to how often milk is bought given bread). Popular algorithms for this include Apriori. These rules help retailers recommend products (e.g. “customers who buy X also buy Y”).
- **Apriori Algorithm:** A classic algorithm for mining frequent itemsets and association rules ¹⁵ ¹⁶ . It works iteratively: first find all frequent 1-itemsets above support threshold, then generate candidate 2-itemsets only from those frequent 1-itemsets, prune infrequent ones, and so on. It uses the “Apriori property”: all subsets of a frequent itemset must also be frequent ¹⁷ . Steps: (1) Identify frequent single items; (2) Generate larger itemsets from frequent ones and prune; (3) Repeat until no new itemsets. Finally, derive association rules from these itemsets that satisfy the confidence threshold. Apriori is simple and widely used, but can be expensive on large datasets due to many iterations.
- **Clustering and its methods:** Clustering groups data points into clusters so that items in a cluster are more similar to each other than to those in other clusters. It is an unsupervised technique. Common methods: *K-means clustering* (partitioning into K clusters by minimizing within-cluster variance ¹⁸ ¹⁹); *Hierarchical clustering* (agglomerative or divisive approaches building a tree of clusters); and *density-based* clustering (e.g. DBSCAN, which finds arbitrary-shaped clusters by density). K-means example: randomly initialize K centroids, assign each point to nearest centroid (Euclidean distance), update centroids as cluster means, and repeat until convergence ¹⁸ ¹⁹ .
- **Bayes’ Theorem in Data Mining:** Bayes’ theorem provides a way to update probabilities based on evidence. In data mining (classification), Naïve Bayes classifier uses Bayes’ theorem to compute the probability that a record belongs to a class given its attribute values. Despite the “naïve” (conditional independence) assumption, it is effective for text classification and other domains. It calculates $P(\text{Class}|\text{Data}) \propto P(\text{Data}|\text{Class}) * P(\text{Class})$, normalizing over all classes. It’s simple, fast, and works well with limited data.

June 2022

Q1(a). What is association rule mining? Explain: (i) support count, (ii) frequent itemset, (iii) association rule, (iv) rule evaluation metrics.

Answer: Association rule mining discovers interesting relationships (associations) among items in large transactional datasets (e.g. “market basket” analysis) ¹⁵ . It identifies patterns such as “people who buy bread and butter also buy milk.” Key concepts:

- **Support count:** The number of transactions in which an itemset appears. For item A, support count = count of transactions containing A. *Support* (as a metric) is this count divided by total transactions ²⁰ .
- **Frequent itemset:** An itemset (set of items) whose support count is above a user-defined minimum support threshold. Only frequent itemsets are considered for rule generation (Apriori prunes infrequent

ones) 16 .

- **Association rule:** An implication of the form $X \Rightarrow Y$, meaning whenever items in X appear in a transaction, items in Y tend to appear as well. X and Y are itemsets, with $X \cap Y = \emptyset$. The rule has *support* (fraction of transactions containing $X \cup Y$) and *confidence* ($\text{support}(X \cup Y) / \text{support}(X)$). An example: {bread, butter} $\Rightarrow$ {milk} with support 2% and confidence 70% means 2% of all transactions contain all three items, and 70% of the transactions with bread and butter also have milk.

- **Rule evaluation metrics:** Common metrics are **support**, **confidence**, and **lift** 20 21 . Support measures how often the rule applies to the dataset. Confidence measures the strength (how often items in Y appear in transactions containing X). Lift measures how much more often X and Y occur together than if they were independent ($\text{lift} = \text{confidence} / (\text{support}(Y))$). A lift > 1 indicates a positive association beyond random chance.

Q2(a). List and describe all four types of OLAP operations.

Answer: OLAP operations allow interactive analysis of multidimensional data. The four primary OLAP operations are:

1. **Roll-up (Aggregation):** Summarize data by climbing up a hierarchy (e.g. from daily sales to monthly sales) or combining dimensions (e.g. summing regions into countries). It increases the level of abstraction.
2. **Drill-down:** The opposite of roll-up; it navigates from high-level summary down to more detailed data (e.g. from quarterly to daily sales). It decreases abstraction, showing finer detail.
3. **Slice:** Select a single dimension value, creating a sub-cube (e.g. fixing the year = 2021 in a 3D cube of Time $\times$ Product $\times$ Location, yielding a 2D slice of that year's data).
4. **Dice:** Select a range of values across multiple dimensions, yielding a smaller cube (e.g. choosing certain years, products, and locations). Essentially, it is a multi-dimensional filter.

These operations help analysts query data cubes flexibly (e.g. slicing by geography or time, rolling up to see totals, and drilling down to see specifics).

Q2(b). Define text mining. List any three applications. What techniques are used to analyze web-usage patterns?

Answer: (See **Dec 2021 Q2(a)** above for text mining definition and applications 7 .)

- **Applications (sample):** Document classification (e.g. spam filtering), sentiment analysis, topic modeling, information retrieval (search), and customer feedback analysis.
- **Web-usage mining techniques:** Web usage mining analyzes clickstream and log data to discover browsing patterns. Techniques include: sessionization (identifying user sessions), path analysis (frequent navigation paths), clustering of user sessions, and sequence pattern mining (e.g. finding common sequences of pages). Machine learning (e.g. Markov models, association rules on page visits) is used. For instance, association rules can identify that users who view page A often next view page B.

Q3(a). Define a data warehouse and list the three types of data warehouses.

Answer: A **data warehouse** (DW) is a subject-oriented, integrated, time-variant, non-volatile collection of

data for decision support ¹ . It consolidates data from operational systems for analysis and reporting. The three common types of data warehouses are:

1. **Enterprise (Centralized) Data Warehouse:** A single warehouse for the entire organization, integrating data from all business units.
2. **Operational Data Store (ODS):** A near-real-time, centralized store that holds current operational data (often used for routine reporting), but typically not heavily historical.
3. **Data Mart:** A subset of the enterprise DW focused on a specific line of business or department (see answer to Dec21 Q5(a) above ¹⁴). Each data mart serves the needs of a particular group, and multiple data marts may roll up into the enterprise warehouse.

Q3(b). *What is data transformation? List and explain the steps involved in data transformation.*

Answer: Data transformation is the process of converting data from source formats into the form needed by the DW. Steps include:

1. **Data cleaning:** Remove errors, handle missing values and noise.
2. **Data formatting:** Convert data types or units (e.g. dates to a standard format, currencies to a common currency).
3. **Derivation and merging:** Compute derived attributes (e.g. $\text{TotalPrice} = \text{Qty} * \text{UnitPrice}$) and merge fields if needed.
4. **Normalization or aggregation:** Collapse detail to summaries if required (e.g. daily to monthly totals), or denormalize for performance.
5. **Code translation:** Map source codes to target domain (e.g. product codes to product names).
6. **Validation:** Ensure transformed data meets quality rules (e.g. referential integrity).

These steps ensure that once loaded, data is consistent and useful for analysis. Transformation is typically done in the ETL process before loading into the warehouse.

Q4(a). *What is data mining? Give four applications of data mining.*

Answer: Data mining is the process of discovering patterns, correlations, or anomalies in large datasets to predict outcomes. It uses statistical, machine learning, and database techniques to extract knowledge (e.g. classification, association, clustering).

Applications:

- **Market Basket Analysis (retail):** Find products often bought together to cross-sell or arrange store layouts.
- **Fraud Detection:** Identify unusual patterns of transactions that may indicate fraud (e.g. banking, insurance).
- **Customer Segmentation:** Group customers by purchasing behavior or demographics (clustering) for targeted marketing.
- **Predictive Maintenance:** Analyze machinery sensor data to predict equipment failures.

Other examples: credit scoring, medical diagnosis (data-driven prediction), recommender systems (e.g. Netflix/movie recommendations), and more.

Q4(b). *What is ETL? Why is ETL needed for a data warehouse? Outline how ETL works in data warehousing.*

Answer: ETL stands for *Extract, Transform, Load*. It is needed to integrate and consolidate data from various operational systems into the data warehouse. Operational sources (transactional databases, flat files, etc.)

often have different formats and quality; ETL cleans and transforms this data into a consistent warehouse schema. ETL ensures that the DW has up-to-date, accurate, integrated data for analysis.

How it works:

- **Extract:** Data is read from source systems (e.g. ERP, CRM, logs). Connections (ODBC, APIs) are used to pull data (often incrementally).
- **Transform:** Extracted data is cleaned (missing values handled, duplicates removed, errors fixed), conformed (standardizing units, formats), and enriched (e.g. adding derived attributes). Business rules (e.g. splitting address fields, mapping codes) are applied. This step may use staging tables or intermediate files.
- **Load:** The transformed data is loaded into the data warehouse schema (dimension and fact tables). This might involve inserting into staging tables first, then updating/inserting into dimension tables (slowly changing dimensions) and fact tables.

ETL is typically scheduled (e.g. nightly) and monitored. Proper ETL design handles dependencies (e.g. load dimensions before facts) and may use bulk operations for efficiency.

Q5. Short notes on any four:

(a) OLTP vs OLAP – differences; (b) Data mart vs data warehouse; (c) Numerical taxonomy; (d) Normalization vs. aggregation.

Answer:

- **OLTP vs OLAP:** OLTP (Online Transaction Processing) systems are for day-to-day operations (e.g. retail sales systems). They are normalized databases optimized for fast inserts/updates and short transactions. OLAP (Online Analytical Processing) systems (like a DW) are optimized for read-intensive analytical queries. OLAP uses denormalized schemas (star/snowflake) and pre-aggregations for fast querying ²² ²³. OLTP queries are simple (single record retrieval/update), whereas OLAP queries often involve complex aggregations over large volumes. OLTP data is current (operational), OLAP data is historical.

- **Data Mart vs Data Warehouse:** A **data warehouse** is a centralized repository for the whole organization. A **data mart** is a smaller, focused repository for a specific department or business area ¹⁴. For example, a Sales data mart contains only sales-related data for the sales team, whereas the enterprise DW contains all corporate data. Data marts speed up queries for their users but may be subsets of the warehouse or sourced separately. Multiple data marts can exist (sometimes integrated via the enterprise DW).

- **Data Normalization vs Aggregation:** *Normalization* is a data modeling process (for OLTP) that eliminates redundancy by splitting data into multiple related tables using primary/foreign keys. In DW context, normalization usually refers to snowflake schemas. *Aggregation* refers to summarizing data (e.g. rolling up daily sales into monthly totals). Aggregated data reduces granularity and data volume. While normalization splits tables, aggregation combines/condenses data. In a DW, one often uses denormalized schemas (star) but may store aggregated (summary) fact tables (called aggregate fact tables) to improve query performance ²⁴.

- **Numerical Taxonomy (if chosen):** In data mining, numerical taxonomy refers to clustering using a numeric measure of similarity. For example, hierarchical clustering builds a tree (taxonomy) by successive mergers. (This term is not common; likely intended as “numerical classification” or clustering.)

Each of these concepts aids in the design and use of data systems.

December 2022

Q1(a). Discuss the following in dimensional modeling (fact tables and dimension tables): (i) fact table, (ii) dimension table, (iii) aggregate fact table, (iv) derived dimension table, (v) conformed dimension.

Answer: In a star schema for a data warehouse:

- **Fact table:** The central table storing *quantitative measures* of a business process (e.g. sales) ²⁵. It has foreign keys to dimension tables and numeric facts (e.g. QuantitySold, AmountSold). Each row represents a transactional event. Example columns: (SaleID, TimeID, ProductID, StoreID, Quantity, SalesAmount). The fact table is often very large.
- **Dimension table:** Tables that describe the *attributes* of dimensions for facts. They contain descriptive fields. For example, a **Product** dimension might have (ProductID, Name, Category, Price). Dimension tables are usually smaller and contain text attributes used for filtering/grouping. They are typically denormalized in a star schema.
- **Aggregate fact table:** A fact table storing **pre-computed summaries** (aggregates) at a higher granularity than the base (leaf) fact table ²⁶. For example, instead of daily sales by store, an aggregate fact might have monthly sales by store. Aggregate tables improve query performance by avoiding on-the-fly aggregations. They are identified by a small set of dimension keys and aggregated measures.
- **Derived dimension table:** A dimension table created from existing dimensions or facts, often to support drill-down. For example, a "Time Quarter" dimension derived from the base "Time" dimension. Alternatively, it can refer to dimensions derived from fact tables (e.g. a list of new products sold). Essentially, it's a dimension not directly coming from operational master data but computed.
- **Conformed dimension:** A dimension table that is shared across multiple fact tables or data marts, and has the same meaning and values throughout the warehouse. For instance, a **Date** dimension used by both Sales and Inventory facts with the same ID and hierarchy. Conforming ensures consistent reporting across different areas.

Q1(b). Explain the components of data warehousing architecture (with examples).

Answer: A typical DW architecture has the following components (often layered) ²⁷:

1. **Data Sources:** Operational databases, flat files, external sources. Data is extracted from these.
2. **ETL / Data Integration:** Tools/processes that extract data from sources, transform (cleanse, format) it, and load it into the warehouse. Example: nightly ETL jobs.
3. **Metadata Repository:** Stores metadata (definitions, mappings, data lineage). Example: A repository listing each table/column and its source.
4. **Data Storage:** The data warehouse itself (fact and dimension tables) ²⁷. Could be on-premise or cloud DB. Also may include data marts and an operational data store (ODS).

5. **Data Warehouse Access Tools / Presentation Layer:** Front-end tools for querying and analysis (e.g. SQL query engines, BI/reporting tools, dashboards, OLAP tools) ²² . Users create reports, OLAP cubes, or ad-hoc queries here.

Example: An organization might load all sales and HR data into a central DW. Analysts use a BI tool (e.g. Tableau) to build dashboards on top of the DW (presentation layer). The metadata guides them in understanding each field (e.g. "SSN" has privacy restrictions).

Q1(c). *What is a decision tree? What are its advantages and disadvantages?*

Answer: (See **Dec 2021 Q1(d)** for decision tree definition ⁶ .)

- **Advantages:** Decision trees are easy to interpret and visualize. They require little data preparation (no need to scale/normalize), can handle both categorical and numerical data, and implicitly perform feature selection by picking the best splits. They can model nonlinear relationships and interactions between features.
- **Disadvantages:** They can easily overfit (learn noise) if not pruned, especially with many levels. Decision trees are unstable: small changes in data can yield different tree structures. They may be biased if some classes dominate. Also, they can be less accurate than other methods (ensemble or statistical) on some tasks. Complex trees can be hard to understand, defeating the simplicity advantage.

Q2(a). *Define data cleaning. Explain how to handle missing values and noisy data in preprocessing.*

Answer: Data cleaning (see **Dec 2021 Q1(c)**) involves correcting or removing errors in data prior to analysis.

- **Missing values:** Can be handled by: (1) **Deletion:** Remove records or attributes with missing values (if few or not needed). (2) **Imputation:** Replace missing entries with estimates (mean/median for numerical fields, mode for categorical). (3) **Prediction:** Use regression or classification to estimate missing values from other attributes. (4) **Flagging:** Mark missing values explicitly if they carry meaning.
- **Noisy data:** Data with errors or outliers. Techniques: (1) **Smoothing:** Binning (group into intervals), regression, or moving averages. (2) **Outlier detection:** Identify outliers (e.g. beyond 3 standard deviations) and either remove or cap them. (3) **Normalization:** Scale data to reduce impact of extreme values.

By cleaning missing and noisy data, the accuracy of data mining models improves and erroneous patterns are avoided.

Q2(b). *Write and explain the K-means algorithm for clustering. How does it work?*

Answer: **K-means clustering** partitions a dataset into **K** clusters by minimizing intra-cluster variance ¹⁸ . The steps are:

1. **Initialize:** Choose K (number of clusters). Randomly select K data points as initial cluster centroids (or use smarter initialization like k-means++).
2. **Assign:** For each data point, compute its distance to each centroid (usually Euclidean distance) and assign the point to the nearest centroid's cluster ¹⁸ .
3. **Update:** After assignment, recompute each cluster's centroid as the mean of all points assigned to it.

4. **Iterate:** Repeat Steps 2–3 until centroids stabilize (or a max number of iterations). Convergence occurs when assignments no longer change significantly.

Example: Suppose $K=2$, with points on a plane. Randomly choose two centroids. Assign each point to the nearer centroid, then recompute centroids. After a few iterations, the algorithm converges: each point belongs to a cluster whose centroid represents the cluster's center.

Working principle: K-means minimizes the **sum of squared distances** between points and their cluster centroids ¹⁸. It's simple and efficient but sensitive to the initial centroid choice and to outliers.

Q3(a). *Why is dimensionality reduction needed for text data? Explain tokenization and the vector-space approach with examples.*

Answer: Text data is typically high-dimensional (each word can be a feature), and sparse. Dimensionality reduction is needed to make text analysis tractable and to highlight important features. By reducing dimensions, we remove noise and reduce computational cost.

- **Tokenization:** The process of breaking text into “tokens” (typically words or terms) ²⁸. For example, the sentence “Data mining is useful” tokenizes to [“Data”, “mining”, “is”, “useful”]. Tokenization is the first step in text processing, enabling each token to be analyzed.
- **Vector-space approach:** After tokenizing, text documents can be represented as vectors in a term (word) space (bag-of-words model). Each dimension corresponds to a term (or token), and the value is typically term frequency or TF-IDF weight. For example, two documents might be represented as vectors indicating counts of each word. This numeric representation allows application of data mining algorithms (e.g. clustering or classification) to text.

By transforming raw text into tokens and then into a vector space, we convert unstructured text into structured numeric data suitable for analysis. Dimensionality reduction methods (like selecting top TF-IDF features, or using Latent Semantic Indexing) can then reduce the number of term dimensions while preserving meaning.

Q3(b). *Enumerate best practices for data warehouse architecture.*

Answer: Best practices include:

- **Use a layered architecture:** Separate staging area, ETL processing, data warehouse, and data marts, so changes in one layer have minimal impact on others.
- **Maintain metadata:** Keep thorough metadata (business and technical) to document source, transformations, and meaning of all data. This aids user understanding and maintenance.
- **Implement conformed dimensions:** Use shared dimensions (e.g. Date, Product) across fact tables/data marts to ensure consistency in reporting.
- **Regular data cleansing:** Apply rigorous cleaning and quality checks during ETL to ensure DW data is reliable.
- **Incremental loading:** Use change data capture and incremental ETL to update the DW efficiently with new data.
- **Design for performance:** Pre-aggregate data, index key columns, and use partitioning on large tables to speed queries.
- **Security and governance:** Enforce access controls, encryption, and audit logging. Have clear data governance policies.

- **Scalability and flexibility:** Architect with scalability in mind (e.g. cloud DW, elastic resources) and allow for adding new data sources or expanding schema as needs change.

Q3(c). Describe the following types of data marts: (i) dependent, (ii) independent.

Answer:

- **Dependent Data Mart:** Built from a central data warehouse. Data is sourced from the enterprise DW (or ODS) and tailored for a specific department or function. It inherits the cleaning and integration of the DW, ensuring consistency. For example, a Sales data mart whose data comes from the EDW's sales tables.

- **Independent Data Mart:** Built directly from source systems, not relying on a central DW. Often created by a specific department for its own needs, without enterprise integration. This can lead to inconsistencies between marts. An example: a marketing team extracts data from CRM and spreadsheets into a stand-alone mart.

Dependent marts provide a subset of the unified DW (consistent but requires a DW); independent marts are quicker to set up (no DW needed) but risk data duplication and integration issues.

Q4(a). Explain the snowflake schema with advantages and disadvantages.

Answer: A **snowflake schema** is a variant of star schema where dimension tables are *normalized* into multiple related tables. For example, a Product dimension may be split into (Product → Category → Department tables).

- **Advantages:**

- Reduced data redundancy (normalized tables avoid repetition).
- Saves storage space (useful if dimension hierarchies are many levels).
- Can reflect changes more easily (updating one level in hierarchy).

- **Disadvantages:**

- More complex queries: Joins between multiple dimension tables can slow query performance.
- Less straightforward to understand visually than a star schema.
- Slightly slower for read-mostly OLAP queries due to extra joins.

In contrast, a star schema uses denormalized dimension tables, which is simpler and often faster for queries ²³.

Q4(b). Describe web mining and its tasks, with features and applications.

Answer: **Web mining** is the application of data mining techniques to extract knowledge from web data (content, structure, and usage). It encompasses:

- **Web content mining:** Extracting knowledge from the content of web pages (text, images, videos) ²⁹. *Example:* classifying web pages or extracting key topics from blog posts.
- **Web structure mining:** Discovering the link structure of the web (e.g. hyperlink graph) ³⁰. *Example:* PageRank and HITS algorithms use structure mining to find authoritative pages.
- **Web usage mining:** Analyzing user behavior data (clickstreams, logs) ³¹. *Example:* discovering frequent navigation paths or segments of users with similar browsing patterns.

Tasks:

- *Content mining tasks*: document clustering, text categorization, information extraction (e.g. finding entities in pages).
- *Structure mining tasks*: hyperlink analysis, mining of page hierarchy or site map.
- *Usage mining tasks*: session identification, pattern discovery (e.g. association rules on page visits), personalization (recommend pages).

Applications: Improving search engines, recommendation systems (e.g. "People also viewed"), web analytics (site optimization), and targeted advertising. For instance, web usage mining can reveal that users who visit page A often proceed to page B, enabling site design improvements or targeted content.

Q5. Short notes on any four:

- (a) Data transformation (in ETL)
- (b) Cloud data warehouse
- (c) Data lake architecture
- (d) Data warehouse automation

Answer:

- **Data Transformation:** (See **June 2022 Q3(b)** above.) In ETL, transformation involves cleaning (e.g. imputing missing values, correcting errors), conforming data (standardizing formats, units), deriving new fields (calculations), and aggregating as needed before load. It ensures DW data is integrated and consistent. For example, converting all dates to YYYY-MM-DD and summing hourly sales into daily totals are transformation steps.

- **Cloud Data Warehouse:** A DW hosted on a cloud platform (e.g. Amazon Redshift, Google BigQuery). **Benefits:** virtually unlimited scalability (elastic storage/computing), reduced maintenance (provider manages hardware), and pay-as-you-go pricing. **Challenges:** concerns over data security and compliance, potential vendor lock-in, and latency if data sources are on-premise. Overall, cloud DW enables cost-effective scaling and integration with cloud analytics tools.

- **Data Lake Architecture:** (See **Dec 2021 Q3(a)**.) A data lake typically uses a **distributed storage** (like HDFS or cloud object store) to hold raw data. Key layers: ingestion (collect data), storage (raw data zone), processing (where schema is applied on read, e.g. Spark jobs), and serving (exposed via SQL engines or data services). Often includes a metadata/catalog layer. It may incorporate a **staging or sandbox** area for preparing data. The architecture ensures all data (structured and unstructured) can be accessed by analytics tools.

- **Data Warehouse Automation:** The use of specialized tools and frameworks to automate DW tasks (ETL development, data modeling, documentation, deployment). Automation can generate ETL code from metadata, auto-create schemas, and schedule processes. Tools (e.g. WhereScape, IRI Voracity) reduce manual coding effort, speed up development, and enforce standards. Automation improves consistency and agility, though it requires careful setup of mappings and models.

June 2023

Q1(a). Discuss the following data preprocessing stages briefly: (i) Data Cleaning, (ii) Data Integration, (iii) Data Reduction, (iv) Data Transformation.

Answer: Data preprocessing prepares raw data for mining:

- **Data Cleaning:** Remove errors, noise and inconsistencies (see previous answers).
- **Data Integration:** Combine data from multiple sources into a coherent dataset. This may involve schema matching, entity resolution (e.g. matching “CustomerID” from two systems) and removing redundancies. For example, integrating sales data from different country databases into one unified table.
- **Data Reduction:** Reduce dataset size while preserving key information ³². Methods include:
- **Dimensionality reduction:** Selecting important features (feature selection) or combining features (e.g. PCA).
- **Aggregation:** Summarizing data (e.g. replacing many sales transactions with monthly totals).
- **Sampling:** Using a representative subset.
- **Numerosity reduction:** Using techniques like histograms or clustering to summarize data. These methods speed up mining by handling fewer data points ³².
- **Data Transformation:** Convert data into appropriate formats or scales. Examples: normalizing numeric values to [0,1], encoding categorical variables (one-hot encoding), or generating new attributes. Also includes discretization (binning continuous data). Transformation ensures different data types are comparable and algorithms work effectively.

Q1(b). Define aggregate fact tables and derived dimension tables.

Answer:

- **Aggregate Fact Table:** A fact table that stores pre-computed summary data (aggregations) at a higher grain than the detailed (leaf) fact table ²⁶. For instance, instead of storing every sale, an aggregate fact may record monthly sales totals by product. These tables accelerate queries that request summarized info. They are often linked to the same dimension tables but at less detailed granularity.

- **Derived Dimension Table:** A dimension that is derived from one or more existing tables, often to simplify queries. For example, a “Geography” dimension could be derived by joining City and Region tables into one. Or a dimension derived from the fact table like “Top Products” dimension listing best-sellers. It essentially reuses existing data to create a new dimension structure.

Q1(c). What are the key challenges of data warehouse design?

Answer: Key challenges include:

- **Heterogeneous data integration:** Combining data from disparate sources (different formats, inconsistent keys) ³³. Ensuring consistent definitions across sources is hard.
- **Data quality:** Dealing with missing, noisy, or inconsistent data. Poor quality can propagate errors into the DW.
- **Performance and scalability:** DW must handle large volumes. Designing schemas and indexing for efficient queries (OLAP queries can be complex). Maintaining fast load and query times as data grows is challenging.
- **Changing requirements:** Business needs evolve. The DW design must accommodate new data sources, new metrics, or restructured dimensions without excessive rework.
- **User acceptance:** Ensuring end users trust the data and find the DW easy to use requires good metadata,

clear definitions, and training.

- **Security/Governance:** Balancing wide access for BI with data privacy and compliance adds complexity.

These challenges require careful upfront planning (robust data modeling, ETL processes, and documentation) and ongoing management.

Q1(d). *Contrast a data lake and a data warehouse.*

Answer: Data lakes and data warehouses both store data, but differ fundamentally ³⁴ :

- **Structure:** A data lake stores raw, unstructured/semi-structured data as-is, without enforcing a schema upfront ¹² . A data warehouse stores structured data, cleaned and formatted for a specific purpose (schema-on-write).

- **Purpose:** Data warehouses serve traditional BI queries on historical data. Data lakes are used for big data analytics and advanced data science (e.g. real-time analytics, machine learning) because they retain all original data (including logs, sensor feeds).

- **Schema:** Data warehouses use schema-on-write (schema defined before loading). Data lakes use schema-on-read (interpret data when reading it).

- **Users:** DWs are mostly used by business analysts and decision-makers. Data lakes are used by data scientists and engineers who need raw data.

- **Flexibility:** Lakes are highly flexible (any data type), but can become "swamps" if unmanaged (unorganized data). Warehouses are more rigid but provide fast, reliable queries.

December 2023

Q1(a). *With a diagram, describe the conceptual architecture of a Hadoop data warehouse.*

Answer: A Hadoop-based data warehouse (often called a *Data Lakehouse*) integrates DW concepts with Hadoop's big data storage/processing:

- **Data Ingestion Layer:** Source systems feed data (structured, unstructured) into Hadoop's distributed file system (HDFS or cloud storage). Tools like Kafka or Sqoop ingest streaming and database data.
- **Storage Layer:** Raw data is stored in HDFS. Hadoop's schema-on-read allows storing JSON, CSV, images, logs, etc., together.
- **Processing Layer:** Data processing frameworks (MapReduce, Hive, Spark) run transformations/ETL and analytics on the stored data.
- **Metadata/Query Layer:** Systems like Hive Metastore or Apache Atlas maintain schema info. SQL-on-Hadoop engines (Hive, Impala, Presto) provide query access.
- **Serving Layer:** BI tools or applications access the processed data via JDBC/ODBC. For example, after cleaning and structuring data via Hive, analysts run SQL queries with BI dashboards.

Diagram (textual description): Source Systems (DBs, logs) → Ingestion (Kafka/Sqoop) → **HDFS Storage** (raw data) ← Metadata (Hive) → Processing (Spark/Hive jobs) → **Structured Tables** (in Hadoop) → Query Layer (HiveQL, Presto) → BI/Analytics.

This architecture allows the scalability of Hadoop with the structured querying of a warehouse. Data in Hadoop can be reprocessed repeatedly, and schema can evolve.

Q1(b). Draw and explain star and snowflake schemas for dimensions (Product, Customer, Time, Location) and fact (Sales-Items) with measures Quantity-sold and Amount-sold.

Answer:

- **Star Schema:** One fact table **Sales-Items** with columns (TimeID, ProductID, CustomerID, LocationID, QuantitySold, AmountSold). Four dimension tables:
 - **Product**(ProductID, Name, Category, etc.)
 - **Customer**(CustomerID, Name, Region, etc.)
 - **Time**(TimeID, Day, Month, Quarter, Year, etc.)
 - **Location**(LocationID, City, State, Country).

Each dimension is directly linked to the fact. Query example: total quantity/amount by Product and Time.

- **Snowflake Schema:** Dimensions are normalized. For example, **Product** might split into **Product**(ProductID, Name, CategoryID) and **Category**(CategoryID, CategoryName). Similarly, **Location** could split into City→State→Country tables. Time could be normalized into separate Year, Month, Day tables. The fact table still contains keys ProductID, CustomerID, etc., but ProductID points to a smaller Product table that points to Category, etc.

In diagrams, star shows fact at center with simple dimension tables; snowflake shows fact → product → category (hierarchy) etc.

Q2(a). Describe four main techniques for eliminating noise (binning, clustering, regression, outliers).

Answer: To clean noisy data:

- **Binning:** Sort data and group into a few “bins.” Replace values with bin mean or median. For example, for numeric attribute, divide sorted data into 5 bins and smooth by group mean to reduce random variation.
- **Clustering:** Identify natural groups; within each cluster, replace noisy values by cluster centroid values. For example, use k-means clustering on the attribute and replace each point by its cluster mean.
- **Regression:** Fit a regression model to predict values. Use the model to replace outliers or noisy values. For instance, in time-series data, fit a trendline and use it to smooth fluctuations.
- **Outlier detection:** Identify outliers (e.g. beyond 3σ from mean or using median absolute deviation) and either remove them or cap them at a threshold. For example, in a height dataset, values 10x average would be flagged as outliers and excluded.

Each technique smooths or corrects data to improve overall quality for mining.

Q2(b). What is clustering? Give an example of the use of cluster analysis.

Answer: *Clustering* is an unsupervised learning task that groups data objects so that those in the same cluster are similar, and those in different clusters are dissimilar. It reveals natural structure in data without pre-labeled classes.

Example: Market segmentation. A retailer collects customer purchase data (age, income, buying habits) and applies clustering (e.g. K-means) to segment customers into groups. The retailer might find clusters like “young budget shoppers” vs “older premium buyers.” These segments help target marketing campaigns: each cluster is treated differently (e.g. promotions geared for each group). Clustering can also be used in image segmentation (grouping pixels) or anomaly detection (small cluster = outliers).

Q3(a). With an example, explain Market Basket Analysis.

Answer: Market Basket Analysis examines customer transaction data to find associations between items. It aims to discover combinations of products that frequently co-occur in purchases.

Example: In a grocery store dataset, suppose 70% of transactions that contain bread also contain butter. Then the rule **Bread** $\Rightarrow$ **Butter** with confidence 0.70 might be identified. If 10% of all transactions have both bread and butter, the support is 10%. This analysis helps stores place related items together or cross-sell (e.g. putting butter near bread). Other examples: at a bookstore, "Customers who buy book A often buy book B." Retailers use these insights for promotions or recommendations. Market basket analysis often uses algorithms like Apriori to find frequent itemsets and then generate such association rules ¹⁵.

Q3(b). Write and explain the Apriori algorithm used to identify frequently occurring elements and associations.

Answer: The **Apriori algorithm** (see [June 2022 Q1\(a\)](#) above ¹⁵) finds all frequent itemsets and derives association rules. It works iteratively using the Apriori property (all subsets of a frequent itemset are also frequent) ³⁵. Steps:

1. **Generate frequent 1-itemsets:** Count all items, keep those meeting minimum support.
2. **Generate candidate k-itemsets:** Using frequent (k-1)-itemsets, combine to form k-itemset candidates. Prune any candidate if any (k-1)-subset is not frequent (by Apriori property).
3. **Count support:** Scan dataset to count support for these candidates, keep those above threshold (frequent k-itemsets) ¹⁶.
4. **Repeat:** Continue steps 2-3 for k=2,3,... until no more frequent itemsets are found ³⁶.

After obtaining all frequent itemsets, generate association rules $X \Rightarrow Y$ from each frequent itemset such that $X \cup Y$ is frequent. Calculate the rule's confidence and retain only rules exceeding a confidence threshold.

Example: In a supermarket, Apriori may find that the itemset {diapers, beer} is frequent. It can then generate the rule *Diapers* $\Rightarrow$ *Beer*. If this rule has high confidence, it suggests customers buying diapers often buy beer, informing promotional strategies.

Q3(c). List and explain any three popular data mining tools.

Answer: Examples of widely used data mining tools include:

- **R:** A free software environment for statistical computing and graphics. It has extensive packages for data mining (e.g. `caret`, `randomForest`) and is popular for research and prototyping.
- **Weka:** An open-source GUI tool (from University of Waikato) providing a suite of ML algorithms (classification, clustering, association) and data preprocessing. Good for learning and quick experiments.
- **RapidMiner:** A commercial (also free edition) platform with a visual workflow interface. It supports ETL, modeling, and evaluation steps with many built-in operators. Good for users who prefer drag-and-drop model building.
- **SAS Enterprise Miner:** A commercial enterprise platform for advanced analytics and data mining, integrated with SAS language capabilities. Used in industry for robust deployments.

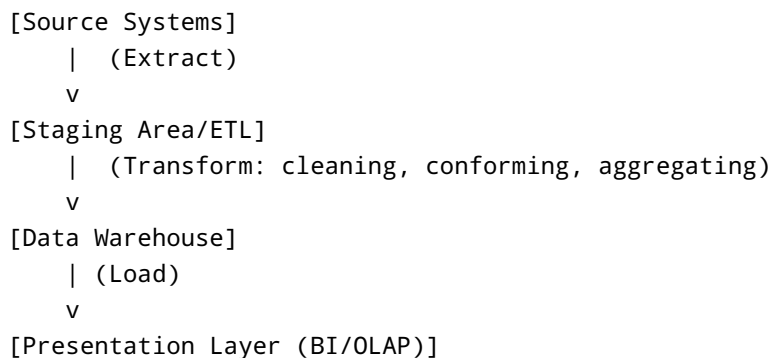
Each tool provides algorithms for classification, clustering, association, etc., and often interactive interfaces. The choice depends on needs (open-source vs commercial, ease of use, scalability).

Q4(a). Discuss ETL and its need. Explain all steps involved in ETL with a suitable diagram.

Answer: **ETL (Extract, Transform, Load)** is essential to populate the data warehouse. It extracts data from

source systems, transforms it into a consistent format, and loads it into the DW. This process is needed because operational data is often disparate and inconsistent – ETL ensures integration, quality, and readiness for analysis.

Steps: (Refer to Dec 2021 Q4(b) above.) Diagrammatically:



1. **Extract:** Connect to various sources (databases, logs, APIs) and retrieve data.
2. **Transform:** Apply data cleaning (handle missing/noise), data integration (combine sources, resolve duplicates), reformatting, deriving new attributes, and business rules.
3. **Load:** Insert transformed data into DW tables. May involve dimension updates (including handling slowly changing dimensions) and fact table insertion.

This process is often orchestrated by scheduling tools and logs progress for error handling. ETL enables the DW to provide a unified, clean view of data for decision support.

Q4(b). List and explain any three key challenges of data warehouse.

Answer: Key challenges (some overlap with Q1(c) of Dec 2022):

- **Data integration:** Combining diverse sources (different formats, semantics) into one model. Aligning schemas and resolving discrepancies is difficult.
- **Performance:** As data size grows, both ETL and query performance must be managed. Designing indexes, aggregations, and efficient queries is challenging for very large warehouses.
- **Data quality:** Garbage in, garbage out. Ensuring clean, accurate data through ETL is hard, especially with legacy or external sources.
- **Changing requirements:** Adapting the DW to new business needs (new metrics, sources) without disrupting existing reports can be complex.
- **Data governance & security:** Ensuring sensitive data is protected, and establishing clear ownership and definitions.

Each of these requires ongoing attention: robust ETL, scalable architecture, and good governance practices.

Q4(c). With reference to Alex Gorelik, explain the following data lake stages: (i) Data Puddle, (ii) Data Pond, (iii) Data Lake, (iv) Data Ocean.

Answer: Alex Gorelik (an industry expert) uses an analogy to describe maturity in big-data storage:

- **Data Puddle:** Early-stage, small isolated datasets (often departmental data marts built on big-data tech). Puddles are standalone and may not integrate (like small puddles scattered).

- **Data Pond:** A collection of related data puddles. Multiple puddles begin to interconnect into a larger repository for an area of business. Still limited in scope.
- **Data Lake:** A centralized large data repository holding raw data from across the enterprise. It's well-managed and supports analytics (the "lake" supports various users diving in).
- **Data Ocean:** The largest scale; an enterprise-wide integrated lake. It covers all data sources and analytics users in the organization. As one "surfs" on the data ocean, the breadth and depth of data and user community are maximal.

As organizations mature, they move from isolated data puddles to a unified data lake/ocean that empowers broad analytics.

Q5. Write short notes on:

- Aggregate fact table and derived dimension table.
- Data swamp.
- Data preprocessing stages.
- Agglomerative hierarchical method.

Answer:

- **(a) Aggregate fact table and derived dimension table:** *Aggregate fact tables* store summarized data at a higher granularity than the base fact table (as described in Dec 2022 Q1(a)). For example, a weekly or monthly summary of sales. *Derived dimension tables* are created by transforming existing data. For example, deriving a "Geography" dimension by combining City and Country tables into one. These help simplify queries or provide customized perspectives.

- **(b) Data Swamp:** A "data swamp" is a derelict data lake. It occurs when data lakes become disorganized and lack governance. In a swamp, data has no clear structure or catalog, making it hard to find or trust information. Essentially, it's a data lake "gone bad." (See [101†L119-L124]: a data lake should be well-organized to avoid this fate.)

- **(c) Data Preprocessing stages:** (See June 2023 Q1(a)). Key stages: **Cleaning** (handle missing values, noise), **Integration** (combine sources, resolve differences), **Transformation** (normalize values, create derived attributes), **Reduction** (sampling, aggregation, feature selection) ³². These prepare the raw data for effective mining or analysis.

- **(d) Agglomerative Hierarchical Method:** A clustering method that builds clusters step by step by merging the two closest clusters iteratively. Starting with each data point as a single cluster, it repeatedly merges the nearest pair of clusters (according to a distance metric) until all points are in one cluster or a stopping condition is reached. This creates a hierarchical tree (dendrogram). It's called "agglomerative" (bottom-up). It is useful for visualizing cluster structure but can be computationally expensive for large datasets.

Sources: Authoritative references and textbooks on data warehousing and mining have been used throughout. Key sources: definitions and concepts from [91†L60-L66] (Data warehouse characteristics), [93†L113-L117] (MOLAP, HOLAP), [101†L160-L168] (metadata, access tools), [104†L194-L202] (star vs snowflake schema differences), [106†L59-L66] (K-means algorithm), [117†L12-L19] (text mining), [121†L22-L30] (Apriori), among others. These provide formal definitions and best-practice details.

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